

Information-optimal illumination for dead-time-limited single-pixel imaging: certificates, water filling, and the jitter-capped ridge

[author block — user decision]

Abstract

A photon-counting detector operated at its deterministic information-optimal flux can forfeit more than half of its recorded information the moment its dead time carries a few percent of timing jitter. This paper asks how hard a real, jittery detector can be driven, and with what illumination — one law answers the first half, one certified design principle the second, and a preregistered campaign seals both. The law: for hidden random hold times of coefficient of variation c_v , we derive an exact missing-information identity and the long-window count-information rate $J_\infty(\rho, c_v) = \rho/[(1+\rho)(1+c_v^2\rho^2)]$, whose optimum is finite for every nonzero hold-time variance, solves the cubic $c_v^2\rho^2(1+2\rho) = 1$, and scales as $2^{-1/3}c_v^{-2/3}$ at small jitter; matching this extensive loss to the deterministic terminal-phase loss of a companion study [1] predicts the smooth crossover $\rho^*(\nu, c_v) \sim (6\nu)^{1/3}(1+12\nu c_v^2)^{-1/3}$, with zero jitter the critical manifold on which the ridge grows as $\nu^{1/3}$ ($\nu = T/\tau$ dead-time slots per exposure). Exact-law peak predictions registered before the refined Monte-Carlo sweep completed were confirmed out-of-sample at both fresh points (3% and 6%), and a pooled two-run continuous-peak fit measures the jitter exponent at -0.658 (1.3% from $-2/3$). The design principle: hand-designed sparse and matched patterns are atoms of one concave experiment-design program whose per-atom information is the exact renewal kernel; the optimum is certified against a frozen structured dictionary under explicit load, dose, and peak constraints, and the program separates into two conjugate resource corners — a photon-budgeted variable-load optimizer (OED-DT) and a time-budgeted ridge-tracking policy with a single global knob (RIDGE-FIXED). A preregistered simulation campaign (M1) with one gate, a common charged pre-scan, six arms, and a mandatory disclosure ledger tests the principle; all outcomes are reported as frozen [M1: primary/secondary verdicts and cross-arm numbers]. The result is an operating handbook: measure the jitter, let the law set the load, let the certificate set the patterns.

1 Introduction

Drive a photon-counting detector to the flux where its count information peaks, and a five-percent timing jitter in its dead time silently forfeits about 55% of that information (Table 1). Single-pixel imaging (SPI) and computational ghost imaging reconstruct a scene from the responses of a single bucket detector to structured illumination [2, 3, 4]; in the photon-starved regime the bucket is a photon-counting device that is blind for a dead time τ after each registered count, and the conventional response is to operate at low flux, keeping the per-slot load $\rho = \lambda\tau$ at the few-percent level and paying in acquisition time. A companion study [1] showed that this convention leaves gains on the table: the exact Fisher information of the integrated nonparalyzable count possesses a principal ridge at the cube-root load $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, and that work mapped when driving a fixed illumination family toward the ridge helps. The trap above is the sharp edge of that map: the deterministic ridge is not where a *real* detector should operate.

Two questions therefore remain open. First, a *deployment* question: how hard can a real detector — whose dead time is a random hold time with some measurable coefficient of variation — actually be driven before the ridge collapses? Second, a *design* question: under a fixed incident-photon and dose budget, which illumination extracts the most information from the dead-time channel, and by how much can a certified optimizer improve on a strong fixed hand design?

One law answers the first question: hidden hold-time randomness adds an *extensive* missing-information term, the long-window optimum becomes finite at the root of an exact cubic, and the deterministic ridge is the zero-jitter critical manifold of a two-parameter crossover (Section 2); its peak predictions, registered before the refined Monte-Carlo sweep existed, were confirmed out-of-sample. One certified design principle answers the second: hand-designed patterns are atoms of a single concave experiment-design program over design measures with the exact renewal kernel, certified against a frozen dictionary, and the program separates cleanly into a photon-budgeted corner and a time-

budgeted corner (Section 3). A preregistered campaign (M1) seals both with one gate and a mandatory disclosure ledger (Sections 4–5).

We formulate a local approximate-design framework for single-pixel illumination in which each pattern’s information is computed from the exact finite-window nonparalyzable renewal count law, optimized under explicit detector-load, dose, per-pixel-exposure, and peak-irradiance constraints and accompanied by generalized-equivalence and exact-rounding certificates. Existing photon-counting optimal experiment design, adaptive or learned SPI, and dead-time optimal-flux studies address neighboring components [5, 6, 7, 8, 9, 10]; we found no prior framework combining these elements for integrated-count single-pixel illumination. The certificate is always stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model.

This work contributes: (i) the random-hold count-information law (Section 2): an exact missing-information identity, the long-window rate with its exact jitter-limited cubic optimum and small-jitter coefficient $2^{-1/3}$, and the matched crossover prediction $(6\nu)^{1/3}(1 + 12\nu c_v^2)^{-1/3}$, verified out-of-sample against preregistered predictions with a pooled jitter-exponent measurement of -0.658 ; (ii) the certified design principle (Section 3): one concave program with the exact renewal kernel, a conservative relaxed Kiefer–Wolfowitz certificate, frozen guards, and the photon-budget versus time-budget conjugacy as its operating doctrine; (iii) a preregistered simulation campaign (M1) whose scenes, atoms, load grid, guards, endpoints, and single gate were frozen in an immutable commit before any confirmatory reconstruction existed, reported as frozen including any negative primary **[M1: campaign outcomes]**; and (iv) an operating handbook for real detectors (Table 5), ending in a bench-testable prediction. All code, frozen preregistration documents, and per-cell evidence bundles are openly released; full derivations, guard machinery, and verification detail are in the supplement.

2 The law: driving a jittery detector

The recovery-time law of the detector determines a scalar count-information kernel; everything else in this paper allocates illumination against that kernel. The companion’s deterministic detector [1] is the zero-jitter critical manifold of the theory below. Proofs at manuscript rigor are given in the supplement.

2.1 Random-hold count information

Generalize the detector so that after each registered event it blocks for a hidden *random* hold time $B_i \sim H$, iid, independent of the Poisson arrivals and of λ , with mean τ and coefficient of variation $c_v = \sigma_B/\tau$ (the companion’s deterministic model is $c_v = 0$). The observed datum remains the integrated registered count N_T over $T = \nu\tau$; the active time L_T , the state path, and the realized holds are hidden from the count-only observer.

Theorem 1 (missing-information identity) *For $\theta = \log \lambda$, the complete detector-path information is $\mathbb{E}[N_T]$, and the information retained by the scalar count is exactly*

$$I_N(\theta; T) = \mathbb{E}[N_T] - \lambda^2 \mathbb{E}[\text{Var}(L_T | N_T)] \quad (1)$$

for every finite T and every iid hidden hold law H independent of λ .

The companion’s terminal-phase identity is the special case in which the uncertainty in L_T is confined to the terminal detector phase. The identity requires amendment if the hold law depends on λ or the scene, if realized holds are observed and used, if holds are serially correlated, or if the counter is paralyzable.

Theorem 2 (long-window rate) *Under standard nonlattice renewal local-limit conditions,*

$$\frac{I_N(\log \lambda; T)}{\nu} \rightarrow J_\infty(\rho, c_v) = \frac{\rho}{(1 + \rho)(1 + c_v^2 \rho^2)}, \quad (2)$$

depending on H only through its mean and variance. The missing-information rate converges to $c_v^2 \rho^3 / [(1 + \rho)(1 + c_v^2 \rho^2)]$: an extensive loss equal, to first order in $c_v^2 \rho^2$, to the heuristic $\lambda^2 \sigma_B^2 \mathbb{E}[N_T]$ — conditioning on the count partially reveals the accumulated hold fluctuation, contributing the factor $(1 + c_v^2 \rho^2)^{-1}$ — and, unlike the deterministic terminal-phase loss, nonvanishing as $\nu \rightarrow \infty$.

Theorem 3 (exact jitter-limited optimum) *For every fixed $c_v > 0$, $J_\infty(\rho, c_v)$ has a unique interior maximum ρ_c solving the cubic*

$$c_v^2 \rho_c^2 (1 + 2\rho_c) = 1, \quad (3)$$

with small-jitter expansion

$$\rho_c = 2^{-1/3} c_v^{-2/3} - \frac{1}{6} + \frac{2^{1/3}}{36} c_v^{2/3} + O(c_v^{4/3}), \quad (4)$$

whose leading coefficient is $2^{-1/3} = 0.79370\dots$

For exponential holds ($c_v = 1$, outside the small-jitter regime) the exact positive root is $\rho_c = 0.6573\dots$; gamma, bounded two-point, and lognormal holds sharing a small c_v share the leading coefficient, with distribution shape entering at the next order.

2.2 The crossover and its universality class

Matching the extensive loss $c_v^2 \rho^2$ against the deterministic terminal-boundary loss $\rho^2/(12\nu)$ in the joint regime $\nu \rightarrow \infty$, $c_v \rightarrow 0$, $c_v^2 \nu = O(1)$ gives

$$\rho_*(\nu, c_v) \sim \left(2c_v^2 + \frac{1}{6\nu}\right)^{-1/3} = (6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}, \quad (5)$$

with scaling variable $x = 12\nu c_v^2$ and crossover at $c_v \asymp (12\nu)^{-1/2}$: the deterministic limit $x \rightarrow 0$ recovers $(6\nu)^{1/3}$ and the jitter-dominated limit recovers $2^{-1/3} c_v^{-2/3}$. An earlier $\min\{(6\nu)^{1/3}, c_0 c_v^{-2/3}\}$ conjecture is only the two-limit envelope of Eq. (5), with an artificial kink, and is superseded. Per the frozen claim discipline, the exact identity and the long-window limit *imply* the jitter exponent $-2/3$; the full finite-window crossover of Eq. (5) is a matched-asymptotic *prediction* until a uniform two-parameter (triangular-array renewal local-limit) proof is complete [\[supplement: uniform crossover proof, or the prediction label is retained\]](#).

The exponent pair $(1/3, -2/3)$ is universal within the regular finite-variance iid hidden-hold class: iid hidden nonnegative holds whose law is independent of λ , with finite nonzero mean and finite variance, forming a regular small-jitter family, under count-only observation and active-start nonparalyzable operation. It is *not* claimed for heavy-tailed (infinite-variance) holds, correlated or drifting holds, holds observed by the electronics, paralyzable detectors, event-time observations, or λ -dependent recovery. The precise critical statement is: zero hold-time variance is the critical manifold on which the optimal load continues to grow with window length; any fixed nonzero hidden hold variance produces a finite long-window optimum. Adjacent renewal-CLT and random-dead-time literature establishes the mean/variance ingredients and task-specific finite-rate optima [11, 12, 13, 14, 10]; we found no source stating the count-only exponent pair or the extensive/terminal crossover.

2.3 Preregistered verification

The law was tested the way the campaigns are run: predictions first, data second. The exact-law column of Table 1, including two out-of-sample rows, was registered before any refined measurement existed (commit 8a627bb). The grid sweep then confirmed the law at both registered fresh points — $c_v = 0.02$: predicted ≈ 10.4 , measured 10.739 (3%); $c_v = 0.2$: predicted ≈ 2.30 , measured 2.153 (6%) — with every in-sample peak within one grid notch and the $\nu = 200$ column consistent including the crossover correction of Eq. (5). A continuous-peak zoom and an independent replication (Colab, 400,000 frames) then measured the exponent itself: the pooled eight-measurement log-log fit gives slope

Table 1: Scalar count information $J(\rho, \nu = 2000, c_v)$ for lognormal hold times: three-stage peak measurement against the exact-law predictions of Theorem 3/Eq. (5), registered *before* the data (docs/R14_PREREGISTERED_PREDICTIONS.md, commit 8a627bb; verdicts at 1d8d7aa, 4d8310d, bd731d5). Grid: jitter_sfi_v2, 100,000 frames, 25-point log grid (spacing 1.157). Zoom: continuous quadratic-interpolated peaks, local run, 150,000 frames. Replication: independent Colab run, 400,000 frames. Both out-of-sample (o.o.s.) grid rows hit ($c_v = 0.02$: 3%; $c_v = 0.2$: 6%); the pooled eight-measurement log-log fit gives slope -0.658 vs. $-2/3$ (1.3%), with the constant consistent with $2^{-1/3} = 0.794$ within 5–10% (the earlier coarse-grid fit 0.72 was a finite-grid artifact) and a reproducible mild $\approx -8\%$ low bias at the end points (open higher-order note). Hardware warning: at the deterministic ridge ($\rho = 22.3$), 5% jitter leaves $J = 0.43$ vs. 0.95 — about 55% of the scalar count information forfeited. J at peak from the grid run.

c_v	peak ρ (grid / zoom / replic.)	exact law	J at peak
0	22.297 / — / —	22.25 (ridge)	0.9452
0.02	10.739 / 9.645 / 9.610	≈ 10.4 (o.o.s.)	0.8731
0.05	5.173 / 5.154 / 5.700	5.69	0.7848
0.1	3.862 / 3.752 / 3.399	3.50	0.6989
0.2	2.153 / 2.106 / 2.051	≈ 2.30 (o.o.s.)	0.5882
0.3	1.607 / — / —	1.63	0.5042

-0.658 , within 1.3% of the predicted $-2/3$, with the constant consistent with $2^{-1/3}$ within 5–10% and a reproducible mild $\approx -8\%$ end-point low bias left as an open higher-order note (candidate: a lognormal third-moment correction). One honest beat from the record: a mid-sweep effective-multiplier hypothesis ($k = 4/3$) was preregistered and refuted by the very next measured point (commits 5e51088, 1a32758). Run specifications, both-run peak tables, and the frozen three-family exponent battery (tests E1–E5) that remains a preregistered extension are given in the supplement.

3 The certified design principle

3.1 One concave program over design measures

Freeze the coarse pre-scan reconstruction at $\hat{x}$ and parameterize the locally estimable image perturbation as $x(\theta) = \hat{x} + B\theta$, with $B \in \mathbb{R}^{n \times r}$ spanning a declared r -dimensional estimable subspace. For a nonnegative illumination atom a the arrival rate is $\lambda(a) = \Phi a^\top \hat{x} + d$ and the load is $\rho(a) = \tau \lambda(a)$. With $J_{\text{exact}}(\rho, \nu)$ the exact per-slot information of the integrated renewal count for $\log \lambda$ (generalized to the random-hold kernel of Theorem 2 when $c_v > 0$) and score direction $q(a) = \Phi B^\top a / \lambda(a)$, the rank-one local information

atom is the chain-rule form

$$\begin{aligned} H(a; \hat{x}) &= \nu J_{\text{exact}}(\rho(a), \nu) q(a) q(a)^\top \\ &= \nu J_{\text{exact}}(\rho(a), \nu) \frac{\Phi^2}{\lambda(a)^2} B^\top a a^\top B. \end{aligned} \quad (6)$$

A raw sensitivity $g \propto \Phi a$ is incomplete unless the factor $\sqrt{\nu J_{\text{exact}}}/\lambda$ is absorbed into it. For a probability measure ξ over the compact feasible atom set $\mathcal{A}$, define $V(\xi) = V_0 + \int_{\mathcal{A}} H(a) d\xi(a)$ with $V_0 \succeq 0$ the frozen pre-scan information, and maximize the local D-criterion

$$\max_{\xi} \log \det V(\xi) \quad \text{s.t. linear resource constraints,} \quad (7)$$

subject to average load, cumulative dose, family allocation, and peak constraints. The objective is concave in ξ ; it is a *local information surrogate* and is not claimed to optimize reconstruction PSNR or any image-domain risk. The estimable subspace is frozen at $r = 200$ (the top eigenbasis of the fixed comparator’s information matrix, fixed before the solve, with a fixed numerical ridge; conventions in the supplement). If every atom obeys a pointwise equal-load constraint the scalar kernel factors out and the geometry reduces to ordinary linear D-optimality [15]; genuine dead-time-aware design therefore requires controlled load variation.

3.2 Certificate and guards

The optimum is certified, not trusted. By concavity, optimality is equivalent to a bounded reduced sensitivity (the constrained Kiefer–Wolfowitz / general equivalence condition for nonlinear models [15, 16]); because a strict per-pixel-dose dual is not yet implemented, the frozen confirmatory quantity is a conservative additive budget-dual bound, `RELAXED_KW_UPPER_BOUND`, required to satisfy $G_{\text{rel}}/r \leq 10^{-3}$ against an exact scan of the frozen dictionary — a learned proposer may warm-start the search, but every frozen run finishes with the exact oracle [17]. Continuous weights are rounded to an exact 972-row design by efficient rounding and deterministic exchange [18, 19, 20], admissible only at exact D-efficiency ≥ 0.95 . Every remaining guard is a frozen constraint, not a tunable proxy (Table 2); any violation is a declared failure state with the fixed comparator as the predeclared fallback, and no threshold is retuned after freeze. The full guard mathematics, the dictionary listing, and the budget accounting are in the supplement.

3.3 The conjugacy doctrine: photon budget versus time budget

The design admits two conjugate resource corners of the same dead-time information map. The gate-carrying

Table 2: The frozen design machinery at a glance (R10/R11/R13; full statements in the supplement).

Element	Frozen rule
Dictionary	all cyclic translates of the 13 R10 support families (scattered $k = 16/32$, solid 4×4 , Lblob, bars, compact-32, ring); weight powers $p \in \{0, 1\}$ only
Budget	incident-signal inequality $\int \rho_s d\xi \leq 0.60$; detected counts dual-reported, never the sole budget
Peak	shape: $w_j \in [\frac{1}{4}, 4]$, $k \geq 16$; physical: $\max a_{ij} \leq 1536$
Dose	per-pixel mean exposure within $\pm 5\%$ of image mean, re-verified post-rounding
Floors	$V_{\text{OED}} \geq 0.5 V_{\text{FIXED}^*}$; $\text{tr}(V^{-1}) \leq 1.05 \text{tr}(V_{\text{FIXED}^*}^{-1})$
Kernel trust	exact-PMF certification; $J_{\text{mean}}/J_{\text{exact}} \geq 0.90$; RQL log-load bias ≤ 0.05 ; $p_{\text{ceil}} \leq 0.05$ per atom, ≤ 0.01 design-weighted
Certificate	$G_{\text{rel}}/r \leq 10^{-3}$ (relaxed KW, exact dictionary scan)
Rounding	exact 972-row design, $\text{Eff}_D \geq 0.95$

OED-DT arm holds the original incident-signal resource as a hard *inequality*,

$$\int \rho_s(a) d\xi(a) \leq 0.60, \quad (8)$$

so a subset of high-value atoms may enter the ridge zone only if compensated by low-load or omitted atoms (R11 §4); detected counts are dual-reported but are never the sole fairness budget, because $\mathbb{E}[N \mid \rho] \simeq \nu\rho/(1 + \rho)$ has vanishing marginal cost at high load. The no-gate `RIDGE-FIXED` arm instead fixes the support and applies one global multiplier per dwell, calibrated on the pre-scan estimate to mean load $\rho_R(\nu)$, followed by a single global safety clip; it uses a deliberately larger incident budget and is therefore descriptive (R11 §2). The exact production ridge is $\rho_R(\nu) = \min \arg \max_{\rho \in \mathcal{R}_{\text{adm}}(\nu)} J_{\text{exact}}(\rho, \nu)$ over the admissible set $\mathcal{R}_{\text{adm}}(\nu) = \{0 < \rho \leq 24 : p_{\text{ceil}} \leq 0.01, J_{\text{mean}}/J_{\text{exact}} \geq 0.90\}$, using the exact numerical ridge rather than the asymptotic cube-root form (R11 §1). Under the incident-photon budget the relevant scalar efficiency is $J_{\text{exact}}(\rho, \nu)/\rho$, which decreases across the admitted palette; a photon-budgeted optimizer therefore rationally spends on lower loads and *avoids* the ridge, while the ridge belongs to the time-limited, power-available corner. This is a resource-regime statement, not a universal prohibition on ridge-zone allocation: sufficiently valuable directions may still receive larger loads in other cells (R13 §8(f)).

3.4 One program behind every optimum

Every true interior optimum of the declared design problem — over load, support scale, and inten-

sity weighting — satisfies one generalized reduced-sensitivity/KKT system of the master program, whose load coordinate is the information-water-filling equation and whose scale and weighting coordinates price conditioning through the V^{-1} geometry (full system in the supplement); this also explains why raising proxy contrast from $p = 1$ to $p = 2$ can lower the true objective. One caveat is kept deliberately: the master design problem places load, support scale, and intensity weighting in one constrained concave measure optimization, and its support atoms satisfy a common reduced-sensitivity condition; but the empirically observed occupancy ($k \approx 32$) and weighting ($p \approx 1$) optima *motivate* these coordinates and are not themselves claimed as continuous KKT theorems — they were selected from discrete grids and judged by image endpoints, not by the master log det objective.

3.5 Why one global knob suffices for RIDGE-FIXED

Fixed flux performs implicit information water filling when bucket brightness and local directional value satisfy the marginal-information alignment condition $\ell_i J'(\kappa u_i) = \beta$ across active directions ($\ell_i = q_i^\top V^{-1} q_i$ the directional value, $u_i = a_i^\top x$ the bucket brightness); its adequacy is measured by a projected KKT residual — not C_u alone — and outside that condition, variable-load OED can improve without any universal bound on the fixed-flux approximation ratio. On non-negative sparse-support scenes the source-induced load ordering naturally tracks directional value, which is why RIDGE-FIXED needs only a single global multiplier, and why the uniform per-pattern servo — which loses by up to ~ 7 dB in the companion’s dev evidence [1] — fails by erasing exactly that alignment. With random holds, J' vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. The full proposition, its residual near-optimality bound, and its failure modes are in the supplement.

4 Preregistered campaign (M1)

M1 is a separate follow-up campaign with its own seed namespace, manifests, and expected-cell ledger; it does not reopen the companion’s frozen studies. It freezes on the R13 audit plus one immutable commit (tag `m1-freeze`); after that commit no scene, atom family, load grid, guard, endpoint, or gate may change, and if the primary fails there is no third redesign.

Geometry and constants. $n = 32^2 = 1024$; $M_{\text{total}} = 1024 = M_{\text{pre}}(52) + M_{\text{main}}(972)$ for every arm; nonparalyzable active start, $\tau = 50$ ns, $d = 0$,

$\sigma_b = 0$; $\bar{\rho} \in \{0.05, 0.60\}$ for the budget-anchored arms; $\nu \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000\}$; five measurement seeds per (image, condition). Optical time $T_{\text{opt}} = M_{\text{total}} \nu \tau$; design-computation wall time is reported separately and never folded into T_{opt} .

Scenes. Twenty-four fresh confirmatory instances (six generator families $\times$ four instances) with seeds $s_{f,r} = 633000 + 100f + r$, plus six development-only instances $632900 + f$. The six committed generators (glyph, chirp, spokes, maze, contour, microtexture) are reused verbatim with their actual names — “USAF” is not used unless a USAF target is generated, and spokes is described as a radial resolution target. No acceptance filter of any kind is applied; generators, images, and ROIs receive SHA256 manifests.

Common pre-scan. Every arm receives and pays for the same 52-pattern scene-independent balanced pre-scan at its own $(\bar{\rho}, \nu)$: 32 paired 4×4 -block rows (support 32, amplitude 32), 16 block rows (amplitude 16), and 4 quadrant rows (amplitude 4), so every pixel receives cumulative amplitude $32 + 16 + 4 = 52$ and the row-mean pattern is exactly the all-ones vector; the mean pre-scan load then equals the declared global load for any sum-normalized scene, with no $\hat{x}$ -dependent row scaling (R13 §2). All 52 counts enter every arm’s final reconstruction; only adaptive arms use the pre-scan to choose subsequent patterns. Fixed arms run 972 main patterns; every arm totals exactly 1024 physical exposures.

Arms. Table 3 lists the six arms. Only OED-DT carries a confirmatory gate; the others are ablations, baselines, or the descriptive ridge-tracking demonstration. The R13 freeze audit additionally mandates MATCH1 as a no-gate arm and a committed selection of the global fixed comparator FIXED* from $\{\text{SCAT32}, \text{LBLOB16}, \text{MATCH1}\}$ by the frozen development rule [M1: committed FIXED* selection record]. Reconstruction (RQL with the frozen analytic λ_{TV} rule and pooled initialization) is identical across arms; the primary comparison is each arm’s own fast curve against the common safe ($\bar{\rho} = 0.05$) reference of its own pattern set.

Endpoints and gate. The frozen Q_{90} time-to-quality machinery transfers verbatim: seed-mean radio-metric PSNR, equal-weight PAVA isotonic curves, a 0.50 dB safe-range informativeness floor, the frozen censoring taxonomy, and a 10,000-replicate nested family-stratified bootstrap. The single primary gate applies to OED-DT only (METHOD_SPEED_PASS): median $S_{\text{gate}} \geq 3$, bootstrap lower bound > 1 , and $\geq 18/24$

Table 3: The six M1 arms. Only OED-DT carries the confirmatory gate; the R13 freeze audit additionally requires MATCH1 and a committed FIXED* selection (see text).

Arm	Role	Gate
OED-DT	variable-load exact-kernel D-optimal design (v3 solver; relaxed-KW certificate vs. the frozen dictionary)	primary
OED-EQLOAD	same optimizer, all atoms at $\bar{\rho}$ (kernel ablation)	none
SCAT16	frozen scattered $k = 16$ (companion bridge)	none
SCAT32	best ladder rung by fixed- dwell median	none
LBLOB16	committed clustered sup- port (strongest fixed base- line; spectral-floor reference)	none
RIDGE-FIXED	dwell-dependent ridge- tracking policy (time-budget demonstration)	none

scenes with $S_j > 1$. Two secondaries cannot rescue a failed primary: a fixed-dwell safe-versus-fast gain (METHOD_FIXED_DWELL_PASS; median ≥ 1 dB at the terminal dwell, lower bound > 0 , $\geq 18/24$ positive) and a design gain of OED-DT over FIXED* at the fast terminal dwell (METHOD_DESIGN_PASS; median ≥ 0.50 dB, family-stratified lower bound > 0 , $\geq 18/24$ positive).

Disclosure ledger. Every cell discloses incident and detected photons (predicted and realized), dose ratios, source/peak-irradiance ratio, the load percentiles ρ_5 , ρ_{50} , ρ_{95} , $\rho_{\max}$, predicted and realized ceiling fractions, and mean J_{exact} ; designed cells add the continuous and exact log det objectives, the exact D-efficiency, A-risk, spectral floor, and the relaxed-KW gap. The full nine-dwell kernel/ridge certification table is frozen before any confirmatory scene is opened.

5 Results

5.1 Primary: OED-DT time-to-quality

[M1: METHOD_SPEED_PASS verdict — median S_{gate} , bootstrap lower bound, count of scenes with $S_j > 1$, and the frozen PASS/FAIL call.]

5.2 Design gain: OED-DT vs. FIXED*

[M1: METHOD_DESIGN_PASS — median design gain (dB) at the fast terminal dwell, family-stratified lower bound, count positive.]

5.3 Fixed-dwell safe-versus-fast gain

[M1: METHOD_FIXED_DWELL_PASS — median terminal-dwell gain (dB), count positive, bootstrap lower bound, per-arm ladder.]

5.4 Secondary arms and design diagnostics

[M1: RIDGE-FIXED operating-point demonstration, OED-EQLOAD kernel ablation, and per-cell certificate/rounding diagnostics — summary here; full tables in the supplement.]

6 Discussion: the operating handbook

The two arms expose conjugate resource regimes of the same dead-time information map. Under a linear incident-photon budget the per-photon efficiency $J_{\text{exact}}(\rho, \nu)/\rho$ falls monotonically, so a budget-constrained design rations photons onto lower loads and *avoids* the ridge; the ridge belongs to time-constrained operation, where power is available and the only scarcity is dwell. Reading the flux ridge and the design allocation as one program clarifies that the large fixed-dwell gains of the companion are an operating-point effect, not evidence that a variable-load optimizer must spend into the ridge.

The jitter-capped ridge is the practical caution. The cube-root ridge is a property of the zero-jitter critical manifold; the moment a detector’s hold time carries a coefficient of variation c_v , the long-window optimum is finite — the exact cubic $c_v^2 \rho^2 (1 + 2\rho) = 1$ of Theorem 3, scaling as $2^{-1/3} c_v^{-2/3}$ — and driving past it is actively harmful: at the deterministic ridge ($\rho = 22.3$), 5% jitter already forfeits about 55% of the scalar count information in the refined sweep (Table 1). A deployment must therefore locate its operating point against the jitter-limited ceiling of the real detector, not the deterministic ridge; the safe conclusion is conditional on measured timing jitter.

Together, the law and the design principle compress into the operating handbook of Table 5: characterize the detector, let the law set the load ceiling, check engagement with the pre-scan, and pick the pattern recipe by which resource is scarce.

Two scope boundaries stand. All results are simulation-based, and every certificate is stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model; an optical extension along the companion’s discussion — a coarse complex-field estimate synthesizing an optical displacement that removes common-mode photons before dead-time loss — remains a natural bench direction [1].

Table 4: Cross-arm summary (descriptive; no new gates beyond the OED-DT primary). All entries fill from the frozen M1 analyzer after tag **m1-freeze**.

Arm	median S_{gate}	design (dB)	fixed-dwell (dB)	mean J_{exact}	incident budget	ceiling frac.	verdict
OED-DT	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
OED-EQLOAD	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
SCAT16	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
SCAT32	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
LBLOB16	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
RIDGE-FIXED	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]

Table 5: Operating handbook for dead-time-limited single-pixel imaging. The engagement index Γ is descriptive (the companion’s computable engagement criterion [1]), never a confirmatory gate.

Step	Rule
1. Characterize	measure τ and the hold-time c_v from time-tagged recovery intervals
2. Set the load	jitter-capped optimum from $c_v^2 \rho^2 (1 + 2\rho) = 1$ (long window) or $(6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}$ (finite window); never operate a jittered detector at the deterministic ridge
3. Check engagement	common 52-pattern pre-scan $\rightarrow$ measured C_u ; $\Gamma = C_u \sqrt{\nu \rho / (1 + \rho)} \gtrsim 1$ marks the photon-limited regime
4. Pick patterns	photon-budgeted: certified OED-DT design (relaxed-KW certificate vs. the frozen dictionary); time-budgeted: RIDGE-FIXED — one global multiplier to $\rho_R(\nu)$ (Section 3.5)
5. Doctrine	photon-constrained $\rightarrow$ stay below the ridge (J_{exact}/ρ falls); time-constrained $\rightarrow$ track the jitter-capped ridge

The law itself, however, is bench-testable now, with no imaging required. For a real single-photon counting module, the hold-time coefficient of variation is directly measurable from time-tagged recovery intervals; the theory then predicts a count-information peak at the root of $c_v^2 \rho^2 (1 + 2\rho) = 1$ rather than at the deterministic ridge — for $c_v = 0.05$, a peak near $\rho \approx 5.7$ instead of ≈ 22 at $\nu = 2000$, with about half the information forfeited if the deterministic ridge is used instead (Table 1). Sweeping source power at fixed dwell and estimating the count information from repeated frames traces $J(\rho)$ directly; this single-detector power sweep is the immediate bench experiment for the line.

7 Reproducibility

The design formulation, guards, and campaign follow four frozen rulings: the dead-time OED formulation and equivalence certificate (R10), the ridge-zone load architecture and incident-budget conjugacy (R11), the M1 freeze audit (R13), and the unification ruling (R14, issue “R14 ruling: unification architecture”), whose theorem hierarchy Sec-

tion 2 follows. Exact-law peak predictions for the refined jitter sweep, including the two out-of-sample rows, were registered before the sweep completed ([docs/R14_PREREGISTERED_PREDICTIONS.md](#), commit [8a627bb](#)) and confirmed in three stages (grid verdict [1d8d7aa](#); local zoom [4d8310d](#); pooled Colab replication [bd731d5](#)), with the intermediate $k = 4/3$ hypothesis preregistered at [5e51088](#) and refuted at [1a32758](#). The campaign freezes at tag **m1-freeze** once the R13 checklist passes outcome-blind self-tests; the companion is frozen at tag **paper1-freeze-v1**. Source-of-record documents for every frozen constant are cited by ruling number in the block comments of the manuscript source; full machinery and proofs are in the supplement ([supplement_m1.tex](#)). All code, frozen pre-registration documents, per-cell manifests, and evidence bundles are available at [\[repo URL wording — user decision\]](#); funding and acknowledgments are [\[funding/acknowledgments — user decision\]](#).

References

- [1] Companion manuscript (paper 1). When high flux helps single-pixel imaging: a contrast–dead-time operating map, 2026. in submission; author block per paper 1 (user decision).
- [2] Marco F. Duarte, Mark A. Davenport, Dharmpal Takhar, Jason N. Laska, Ting Sun, Kevin F. Kelly, and Richard G. Baraniuk. Single-pixel imaging via compressive sampling. *IEEE Signal Processing Magazine*, 25(2):83–91, 2008.
- [3] Jeffrey H. Shapiro. Computational ghost imaging. *Physical Review A*, 78(6):061802, 2008.
- [4] Matthew P. Edgar, Graham M. Gibson, and Miles J. Padgett. Principles and prospects for single-pixel imaging. *Nature Photonics*, 13(1):13–20, 2019.
- [5] Feng et al. Optimal illumination pattern design for single-pixel imaging, 2018. arXiv:1806.01340; authors/title VERIFY at submission.
- [6] Ma, Xu, and Maleki. Sensing-matrix design for generalized nonlinear inverse problems, 2021.

- arXiv:2111.03237; authors/title VERIFY at submission.
- [7] Ahmed Attia, Alen Alexanderian, and Arvind K. Saibaba. Goal-oriented optimal design of experiments for large-scale Bayesian linear inverse problems. *Inverse Problems*, 34(9):095009, 2018.
 - [8] Martin Burger, Andreas Hauptmann, Tapio Helin, et al. Sequentially optimized projections in X-ray imaging. *Inverse Problems*, 2021. authors/volume/pages VERIFY at submission.
 - [9] Ery Arias-Castro, Emmanuel J. Candès, and Mark A. Davenport. On the fundamental limits of adaptive sensing. *IEEE Transactions on Information Theory*, 59(1):472–481, 2013. arXiv:1111.4646.
 - [10] Frederic J. N. Jorgensen and Steven G. Johnson. Fundamental bounds and efficient estimation for dead-time-constrained event detection, with application to single-photon lidar, 2026. arXiv:2605.23210.
 - [11] Jörg W. Müller. Dead-time problems. *Nuclear Instruments and Methods*, 112(1–2):47–57, 1973.
 - [12] Daniel F. Yu and Jeffrey A. Fessler. Mean and variance of single photon counting with deadtime. *Physics in Medicine and Biology*, 45(7):2043–2056, 2000.
 - [13] Fredrik Grönberg, Mats Danielsson, and Martin Sjölin. Count statistics of nonparalyzable photon-counting detectors with nonzero pulse length. *Medical Physics*, 45(8):3800–3811, 2018.
 - [14] Robert E. Alvarez. Signal to noise ratio of energy selective x-ray photon counting systems with pileup. *Medical Physics*, 41(11):111909, 2014.
 - [15] Jack Kiefer. Optimum experimental designs. *Journal of the Royal Statistical Society: Series B*, 1959. volume/pages VERIFY at submission.
 - [16] L. V. White. An extension of the general equivalence theorem to nonlinear models. *Biometrika*, 60(2):345–348, 1973.
 - [17] Martin Jaggi. Revisiting Frank–Wolfe: Projection-free sparse convex optimization. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*, volume 28 of *PMLR*, pages 427–435, 2013.
 - [18] Friedrich Pukelsheim and Sabine Rieder. Efficient rounding of approximate designs. *Biometrika*, 79(4):763–770, 1992.
 - [19] Aleksandar Nikolov, Mohit Singh, and Uthaipon Tantipongpipat. Proportional volume sampling and approximation algorithms for A-optimal design. *Mathematics of Operations Research*, 2022. volume/pages VERIFY at submission.
 - [20] Guillaume Sagnol and Radoslav Harman. Computing exact D-optimal designs by mixed integer second-order cone programming. *The Annals of Statistics*, 43(5):2198–2224, 2015. arXiv:1307.4953.